Elementary Mathematics

(प्रारंभिक गणित)

MAT-101 · Credit 2 (1+1)

Model Paper 1 with Solutions (syllabus based)

मॉडल पेपर 1 - पाठ्यक्रम आधारित, स्टेप-बाय-स्टेप हल

Time	Maximum Marks	Sections
2 Hours	70	A: any 5 × 5 = 25 · B: any 3 × 15 = 45

Not an official university paper; prepared on the syllabus and previous paper pattern. (विश्वविद्यालय का आधिकारिक पेपर नहीं)

Aditya Kumar

M.Sc. (Ag.) Soil Science, Gold Medalist

SECTION A

Short Answer Questions

लघु उत्तरीय प्रश्न

Attempt any five questions (all 8 solved here). Each carries 5 marks.

कोई पाँच प्रश्न कीजिए (यहाँ सभी 8 हल)। प्रत्येक 5 अंक।

5 × 5 = 25 Marks

A.1 **Find the coordinates of the point which divides the join of (2, 3) and (5, -3) externally in the ratio 2 : 1.** **5 Marks**

(2, 3) और (5, -3) को मिलाने वाली रेखा को 2 : 1 में बाह्य विभाजित करने वाले बिंदु के निर्देशांक ज्ञात कीजिए।

Solution (हल)

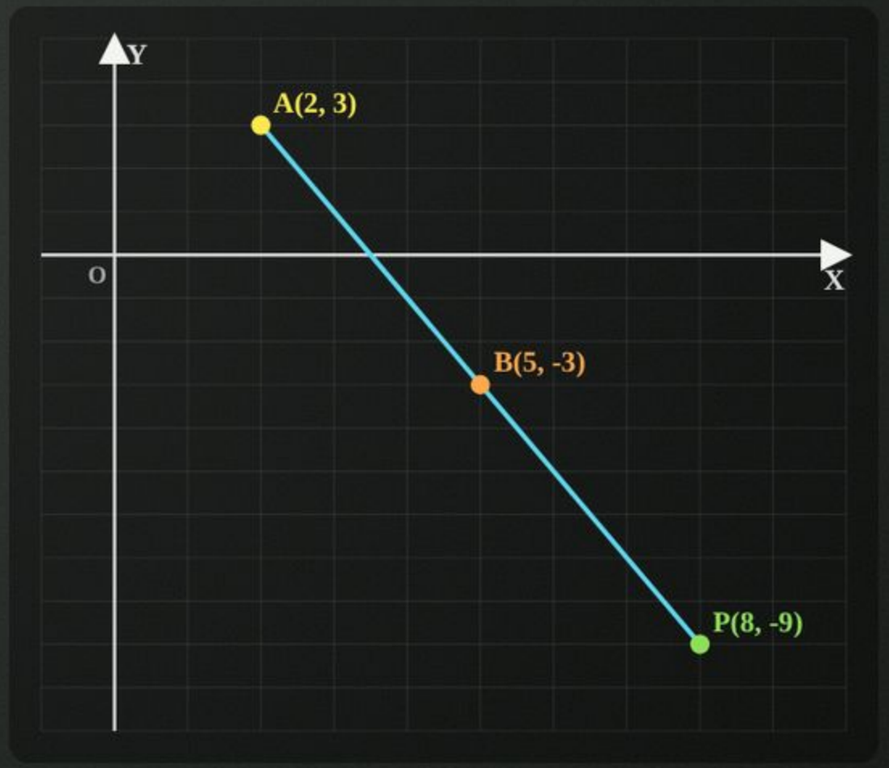
External division: $P = \left(\frac{m x_2 - n x_1}{m - n} , \frac{m y_2 - n y_1}{m - n} \right)$ बाह्य विभाजन सूत्र

$m : n = 2 : 1$, $(x_1, y_1) = (2, 3)$, $(x_2, y_2) = (5, -3)$

$$x = \frac{2(5) - 1(2)}{2 - 1} = 8$$

$$y = \frac{2(-3) - 1(3)}{2 - 1} = -9$$

Required point = **P (8, -9)**



P lies outside AB (external division)

P, AB के बाहर स्थित

A.2 **Find the angle between the lines $y = \sqrt{3}x + 5$ and $\sqrt{3}y = x - 6$.** **5 Marks**

रेखाओं $y = \sqrt{3}x + 5$ और $\sqrt{3}y = x - 6$ के बीच का कोण ज्ञात कीजिए।

Solution (हल)

Line 1: $y = \sqrt{3}x + 5 \Rightarrow m_1 = \sqrt{3}$

Line 2: $y = \frac{1}{\sqrt{3}}x - \frac{6}{\sqrt{3}} \Rightarrow m_2 = \frac{1}{\sqrt{3}}$

$\tan\theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$ | दो रेखाओं के बीच कोण

$\tan\theta = \left| \frac{\sqrt{3} - 1/\sqrt{3}}{1 + 1} \right| = \frac{2/\sqrt{3}}{2} = \frac{1}{\sqrt{3}}$

$\theta =$ **30°** (obtuse angle = 150°)

Remember (याद रखें)

Parallel: $m_1 = m_2$

Perpendicular: $m_1 m_2 = -1$

Slope = $\tan\theta$ (angle with x-axis)

A.3 **Find the centre and radius of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$.** **5 Marks**

वृत्त $x^2 + y^2 - 4x + 6y - 12 = 0$ का केंद्र व त्रिज्या ज्ञात कीजिए।

Solution (हल)

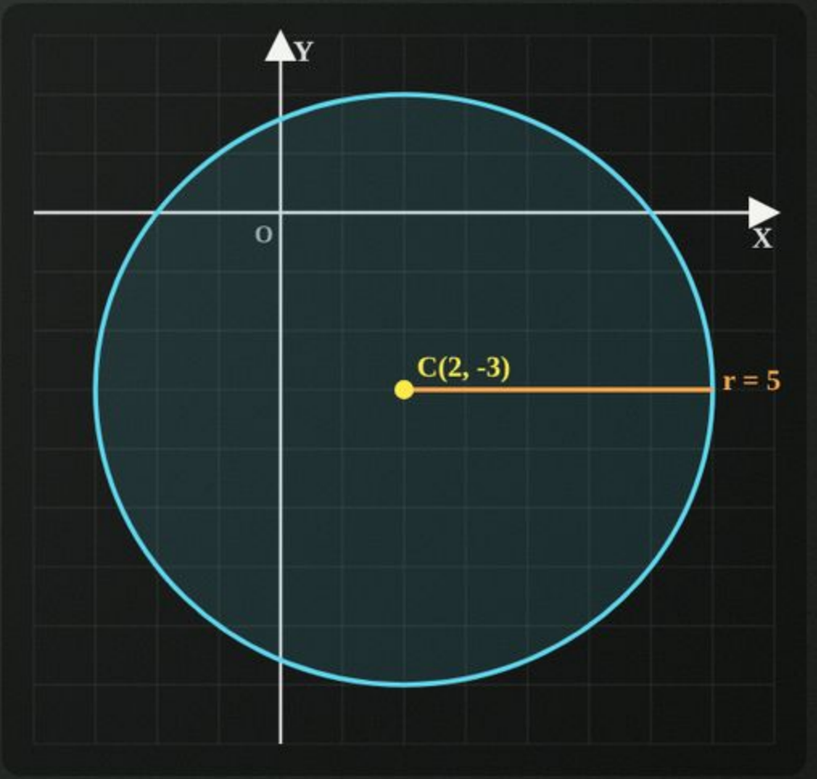
General form: $x^2 + y^2 + 2gx + 2fy + c = 0$; centre $(-g, -f)$, $r = \sqrt{(g^2 + f^2 - c)}$

व्यापक समीकरण

Comparing: $2g = -4 \Rightarrow g = -2$; $2f = 6 \Rightarrow f = 3$; $c = -12$

Centre = $(-g, -f)$ = **(2, -3)**

$r = \sqrt{(4 + 9 + 12)} = \sqrt{25}$ = **5 units**



A.4 **Find the derivative of sin x from first principle.** **5 Marks**

प्रथम सिद्धांत से sin x का अवकलज ज्ञात कीजिए।

Solution (हल)

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$ प्रथम सिद्धांत

$$= \lim_{h \to 0} \frac{\sin(x+h) - \sin x}{h} = \lim_{h \to 0} \frac{2\cos(x + h/2) \cdot \sin(h/2)}{h}$$ [sin C - sin D = 2 cos ((C+D)/2) sin ((C-D)/2)]

$$= \lim_{h \to 0} \cos\left(x + \frac{h}{2}\right) \times \lim_{h \to 0} \frac{\sin(h/2)}{h/2}$$

$$= \cos x \times 1 \quad [\text{since } \lim_{\theta \to 0} \frac{\sin\theta}{\theta} = 1]$$ मानक सीमा

$$\frac{d}{dx}(\sin x) = \boxed{\cos x}$$

A.5 Differentiate x^x with respect to x.

5 Marks

x^x का x के सापेक्ष अवकलन कीजिए।

Solution (हल)

Let $y = x^x$. Taking log on both sides (logarithmic differentiation): लघुगणकीय अवकलन

$$\log y = x \log x$$

$$\frac{1}{y}\frac{dy}{dx} = x \cdot \frac{1}{x} + \log x \cdot 1 = 1 + \log x$$

$$\frac{dy}{dx} = y\,(1 + \log x) = \boxed{x^x\,(1 + \log x)}$$

When to use (कब प्रयोग करें)

When variable is in both base and power, e.g. x^x, $(\sin x)^x$.

जब आधार व घात दोनों में चर हो।

A.6 **Evaluate:** $\lim_{x \to 2} \dfrac{x^3 - 8}{x - 2}$ **5 Marks**

मान ज्ञात कीजिए: $x \to 2$ पर $(x^3 - 8)/(x - 2)$ की सीमा

Solution (हल)

Direct substitution gives 0/0 (indeterminate form); so factorise. 0/0 रूप, गुणनखंड करें

$x^3 - 8 = (x - 2)(x^2 + 2x + 4)$

$$\lim_{x \to 2} \frac{(x - 2)(x^2 + 2x + 4)}{x - 2} = \lim_{x \to 2} (x^2 + 2x + 4)$$

$= 4 + 4 + 4 =$ **12**

Shortcut: $\lim_{x \to a} \dfrac{x^n - a^n}{x - a} = n\,a^{n-1} = 3 \times 2^2 = 12$ ✓ सूत्र से जाँच

A.7 Evaluate: $\int x\, e^x\, dx$

5 Marks

मान ज्ञात कीजिए: $\int x\, e^x\, dx$

Solution (हल)

Integration by parts: $\int u\, v\, dx = u \int v\, dx - \int [\, u' \int v\, dx\,]\, dx$ खंडशः समाकलन

Take $u = x$ (algebraic), $v = e^x$ (exponential) - ILATE rule ILATE नियम

$I = x\, e^x - \int 1 \cdot e^x\, dx = x\, e^x - e^x + C$

$I =$ **$e^x\,(x - 1) + C$**

ILATE (क्रम)

Inverse trig → Log → Algebraic → Trig → Exponential: choose first as u.

जो पहले आए उसे u लें।

A.8 **If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, show that $A^2 - 5A = 2I$.** **5 Marks**

यदि A दिया है, तो दिखाइए कि $A^2 - 5A = 2I$

Solution (हल)

$$A^2 = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1+6 & 2+8 \\ 3+12 & 6+16 \end{pmatrix} = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix}$$

$$5A = \begin{pmatrix} 5 & 10 \\ 15 & 20 \end{pmatrix}$$

$$A^2 - 5A = \begin{pmatrix} 7-5 & 10-10 \\ 15-15 & 22-20 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \boxed{2I}$$ Hence shown.

SECTION B

Long Answer Questions

दीर्घ उत्तरीय प्रश्न

Attempt any three questions (all 5 solved here). Each question carries 15 marks.

कोई तीन प्रश्न कीजिए (यहाँ सभी 5 हल)। प्रत्येक प्रश्न 15 अंक का है।

3 × 15 = 45 Marks

B.1 **(i) Find the equation of the circle passing through (0, 0), (2, 0) and (0, 4). (ii) Find the equations of the bisectors of the angles between $3x + 4y - 7 = 0$ and $12x - 5y + 8 = 0$. (iii) Show that $y = x + 5\sqrt{2}$ touches the circle $x^2 + y^2 = 25$.** **15 Marks**

(i) तीन बिंदुओं से वृत्त (ii) कोण समद्विभाजक (iii) स्पर्शता की शर्त

Part (i) (भाग (i))

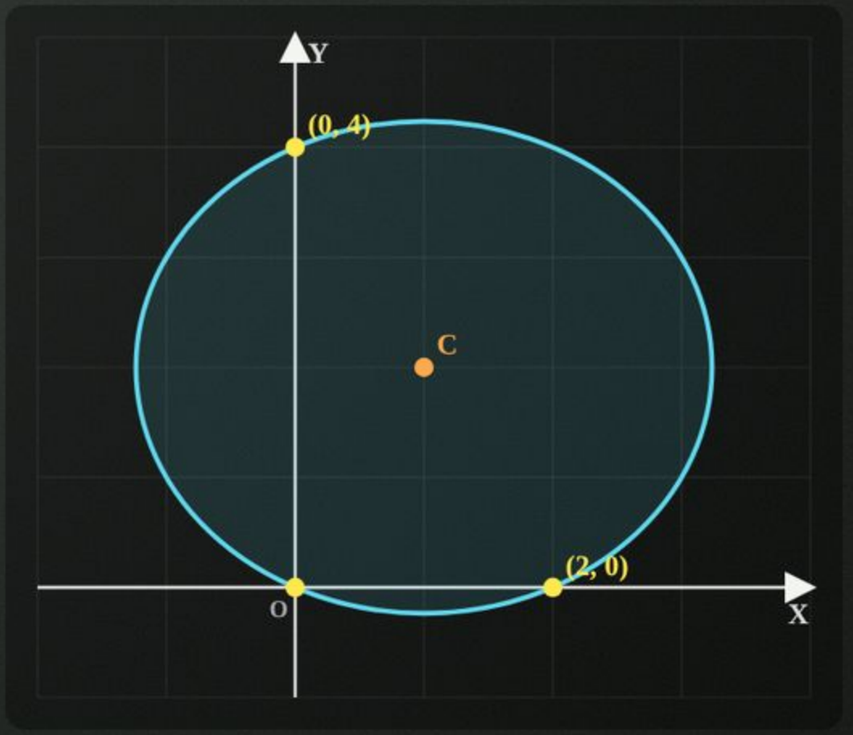
Let $x^2 + y^2 + 2gx + 2fy + c = 0$ व्यापक समीकरण

(0, 0): $c = 0$

(2, 0): $4 + 4g = 0 \Rightarrow g = -1$

(0, 4): $16 + 8f = 0 \Rightarrow f = -2$

Circle: **$x^2 + y^2 - 2x - 4y = 0$** centre (1, 2), $r = \sqrt{5}$



Circle through three points

तीन बिंदुओं से जाने वाला वृत्त

B.1 (ii) Angle bisectors (iii) Condition of tangency

(ii) कोण समद्विभाजक (iii) स्पर्शता

contd.

Part (ii) (भाग (ii))

Bisectors: $\frac{a_1x + b_1y + c_1}{\sqrt{(a_1^2 + b_1^2)}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{(a_2^2 + b_2^2)}}$ समद्विभाजक सूत्र

$\frac{3x + 4y - 7}{5} = \pm \frac{12x - 5y + 8}{13} \Rightarrow 13(3x + 4y - 7) = \pm 5(12x - 5y + 8)$

(+): $39x + 52y - 91 = 60x - 25y + 40 \Rightarrow$ **$21x - 77y + 131 = 0$**

(-): $39x + 52y - 91 = -60x + 25y - 40 \Rightarrow 99x + 27y - 51 = 0 \Rightarrow$ **$33x + 9y - 17 = 0$**

Part (iii) (भाग (iii))

Line $y = mx + c$ touches $x^2 + y^2 = a^2$ if $c^2 = a^2(1 + m^2)$ स्पर्शता की शर्त

Here $m = 1$, $c = 5\sqrt{2}$, $a^2 = 25$: $c^2 = 50$; $a^2(1 + m^2) = 25 \times 2 = 50$

$c^2 = a^2(1 + m^2)$, hence the line touches the circle.

B.2 **Differentiate w.r.t. x: (i) $\frac{x^2+1}{x-1}$ (ii) $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ (iii) $(\sin x)^{\cos x}$** **15 Marks**

x के सापेक्ष अवकलन कीजिए: (i) भागफल (ii) प्रतिस्थापन (iii) लघुगणकीय

Part (i) (भाग (i))

Quotient rule: $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\,u' - u\,v'}{v^2}$ भागफल नियम

$$= \frac{(x-1)(2x) - (x^2+1)(1)}{(x-1)^2}$$

$$= \frac{2x^2 - 2x - x^2 - 1}{(x-1)^2} = \boxed{\frac{x^2 - 2x - 1}{(x-1)^2}}$$

Part (ii) (भाग (ii))

Put $x = \tan\theta \Rightarrow \theta = \tan^{-1} x$ प्रतिस्थापन

$$\frac{2\tan\theta}{1-\tan^2\theta} = \tan 2\theta \Rightarrow y = \tan^{-1}(\tan 2\theta) = 2\theta$$

$$y = 2\tan^{-1} x \Rightarrow \frac{dy}{dx} = \boxed{\frac{2}{1+x^2}}$$

Part (iii) (भाग (iii))

(iii) $y = (\sin x)^{\cos x} \Rightarrow \log y = \cos x \cdot \log \sin x \Rightarrow \frac{1}{y}\frac{dy}{dx} = \cos x \cdot \frac{\cos x}{\sin x} + \log \sin x \cdot (-\sin x)$

$$\frac{dy}{dx} = \boxed{(\sin x)^{\cos x}\,[\,\cos x \cdot \cot x - \sin x \cdot \log \sin x\,]}$$

B.3 **(i) A farmer has 400 m of fencing to enclose a rectangular field. Find the dimensions that give the maximum area. (ii) Find the maximum and minimum values of $y = x^3 - 6x^2 + 9x + 1$.** **15 Marks**

(i) 400 मी बाड़ से अधिकतम क्षेत्रफल वाला आयताकार खेत (ii) उच्चिष्ठ व निम्निष्ठ मान

Part (i) (भाग (i))

Let length = x, breadth = y; $2(x + y) = 400 \Rightarrow y = 200 - x$ परिमाप

Area $A = x(200 - x) = 200x - x^2$

$\frac{dA}{dx} = 200 - 2x = 0 \Rightarrow x = 100$

$\frac{d^2A}{dx^2} = -2 < 0 \Rightarrow$ maximum

100 m × 100 m (square), A = 10000 m² = 1 ha

Part (ii) (भाग (ii))

$\frac{dy}{dx} = 3x^2 - 12x + 9 = 3(x - 1)(x - 3) = 0$

$x = 1$ or $x = 3$; $y'' = 6x - 12$

$x = 1$: $y'' = -6 < 0 \Rightarrow$ max, $y = 1 - 6 + 9 + 1 =$ **5**

$x = 3$: $y'' = 6 > 0 \Rightarrow$ min, $y = 27 - 54 + 27 + 1 =$ **1**



Agri use: maxima-minima help in finding the most profitable input level, e.g., fertiliser dose for maximum yield.

कृषि उपयोग: अधिकतम उपज हेतु उर्वरक की उचित मात्रा ज्ञात करना।

B.4 **Evaluate: (i) $\int x^2 \log x \, dx$ (ii) $\int \frac{2x}{1+x^2} dx$ (iii) $\int_0^{\pi/2} \sin^2 x \, dx$ (iv) Area under $y = x^2$ from $x = 0$ to $x = 3$** **15 Marks**

समाकलन व वक्र के नीचे क्षेत्रफल

Parts (i), (ii) (भाग (i), (ii))

(i) By parts, $u = \log x$, $v = x^2$ (ILATE) खंडशः

$= \log x \cdot \frac{x^3}{3} - \int \frac{1}{x} \cdot \frac{x^3}{3} dx = \boxed{\frac{x^3}{3} \log x - \frac{x^3}{9} + C}$

(ii) Put $1 + x^2 = t \Rightarrow 2x \, dx = dt$ प्रतिस्थापन

$\int dt/t = \log t + C = \boxed{\log (1 + x^2) + C}$

Parts (iii), (iv) (भाग (iii), (iv))

(iii) $\sin^2 x = \frac{1 - \cos 2x}{2}$

$= \frac{1}{2} [x - \frac{\sin 2x}{2}]_0^{\pi/2} = \frac{1}{2} \times \frac{\pi}{2} = \boxed{\pi/4}$

(iv) Area $= \int_0^3 x^2 \, dx = [\frac{x^3}{3}]_0^3 = 9 - 0 = \boxed{9 \text{ sq. units}}$

Area under curve $y = f(x)$ between $x = a$ and $x = b$ is $\int_a^b f(x) \, dx$ (shaded region).

वक्र के नीचे क्षेत्रफल = a से b तक f(x) का निश्चित समाकल।



B.5 **(i) Find the inverse of** $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$ **(ii) Prove that** $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a - b)(b - c)(c - a)$

15 Marks

(i) 3×3 आव्यूह का व्युत्क्रम (ii) सारणिक के गुणधर्म

Part (i) (भाग (i))

$|A| = 1(0 - 24) - 2(0 - 20) + 3(0 - 5) = -24 + 40 - 15 = 1 \neq 0$, so A^{-1} exists.

Cofactors: $C_{11} = -24$, $C_{12} = 20$, $C_{13} = -5$; $C_{21} = 18$, $C_{22} = -15$, $C_{23} = 4$; $C_{31} = 5$, $C_{32} = -4$, $C_{33} = 1$ सहखंड

$$\text{adj } A = (\text{cofactor matrix})^T = \begin{pmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{pmatrix} \Rightarrow A^{-1} = \frac{\text{adj } A}{|A|} = \begin{pmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{pmatrix}$$

Check: first row of A × first column of $A^{-1} = -24 + 40 - 15 = 1$ ✓ जाँच

B.5 (ii) Vandermonde determinant

(ii) वान्डरमोंड सारणिक

contd.

Solution (हल)

$$\Delta = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$R_1 \to R_1 - R_2,\ R_2 \to R_2 - R_3$:

$$\Delta = \begin{vmatrix} 0 & a-b & a^2-b^2 \\ 0 & b-c & b^2-c^2 \\ 1 & c & c^2 \end{vmatrix}$$

Take $(a - b)$ from R_1 and $(b - c)$ from R_2:

$$\Delta = (a-b)(b-c) \begin{vmatrix} 0 & 1 & a+b \\ 0 & 1 & b+c \\ 1 & c & c^2 \end{vmatrix}$$

Expand along C_1: $\Delta = (a - b)(b - c) \times 1 \times [\ (b + c) - (a + b)\ ] = (a - b)(b - c)(c - a)$

$\Delta = (a - b)(b - c)(c - a)$ Hence proved.